8-Puzzle Problem -Weighted Best-First Search and Heuristics

ACOL333: Principles of Artificial Intelligence - Assignment 1
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12th October 2026

Abstract

We explore informed search strategies for solving the given 8-puzzle problem under an evaluation function $f(n) = g(n) + W \cdot h(n)$. We analyse three heuristics: Misplaced Tiles (h_1),Manhattan Distance (h_2) and Linear Conflict (h_3), with (h_1) and (h_2) being our baselines, furthermore proving that h_3 dominates both h_1 and h_2 while preserving admissibility and consistency. Through benchmarking over 38 states of varying optimal solution depth ($4 \leq d \leq 31$) and heuristic weights ($W \in \{0, 0.5, 1, 1.5, 2, 3, 5\}$ plus greedy best-first), performance would be evaluated based on four foundational AI search criteria: completeness, cost optimality, time complexity, and space complexity. Our findings demonstrate the exponential pruning advantages of heuristic dominance and characterize the bounded-suboptimality trade-offs introduced by $W > 1$.

1 Introduction

The 8-puzzle consists of a 3×3 grid containing eight numbered sliding tiles and a single blank space. The objective is to achieve the standard goal state from an initial configuration:

$$\begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$$

where 0 denotes the empty slot. Although the state space is compact ($9!/2 = 181{,}440$ reachable states), uninformed search (such as Breadth-First Search or Uniform Cost Search) exhibits exponential memory and time demands as the shortest path distance d increases.

To overcome these computational bottlenecks, we generalize the node evaluation function to:

$$f(n) = g(n) + W \cdot h(n) \tag{1}$$

where $g(n)$ is the exact path cost from the start node to node n, $h(n)$ is a heuristic estimate of the cost from n to the goal and $W \geq 0$ is a scalar weight. When $W = 0$, the algorithm reduces to Uniform Cost Search (Dijkstra's Algorithm). When $W = 1$, it corresponds to standard A^* search. For $W > 1$, it executes Weighted A^* (WA^*), trading bounded solution optimality for reduced node expansion counts. Greedy best-first search, which ranks nodes by $h(n)$ alone, is the limiting case $W \to \infty$.

2 Heuristic Formulations and Theoretical Analysis

Let S be a board state. For each tile $t \in \{1, \ldots, 8\}$, let (r_t, c_t) denote its current coordinates and (r_t^*, c_t^*) denote its coordinates in the goal state.

2.1 Heuristic Definitions

1. Misplaced Tiles (h_1): Counts the number of non-blank tiles that do not occupy their goal position:

$$h_1(S) = \sum_{t=1}^{8} \mathbf{1}\{(r_t, c_t) \neq (r_t^*, c_t^*)\} \tag{2}$$

2. Manhattan Distance (h_2): Measures the sum of grid distances each tile must traverse to reach its goal slot, ignoring inter-tile collisions:

$$h_2(S) = \sum_{t=1}^{8} (|r_t - r_t^*| + |c_t - c_t^*|) \tag{3}$$

3. Linear Conflict (h_3): Extends Manhattan Distance by accounting for tile interference within the same row or column. Let's say two tiles t_j and t_k are in a *linear conflict* if they reside in the same row (or column), both have their goal destinations in that same row (or column), but their current relative order is inverted with respect to their goal order:

$$h_3(S) = h_2(S) + 2 \cdot LC(S) \tag{4}$$

where $LC(S)$ is the number of tiles in the line minus the length of the longest increasing subsequence of their goal positions, summed over all rows and columns. For example, with goal order 3,2,1 there are 3 conflicting pairs, but only 2 tiles need to leave the line. Counting pairs would add 6 instead of 4 and overestimate; over the full state space,pair counting exceeds h^* on 7 states, whereas our formulation never does.

2.2 Theoretical Properties

Theorem 1 (Admissibility and Consistency of h_3)**.** *The Linear Conflict heuristic h_3 is both admissible and consistent.*

Proof. Consider a row whose goal-row tiles are out of order. Tiles in the same row cannot pass each

other, so every tile outside a longest correctly ordered subsequence must vacate the row into an adjacent row, let the others slide past, and return. Each such detour requires at least two vertical moves (one out, one back) that are not accounted for in the Manhattan distance, since that tile's vertical distance to its goal row is zero; the same for columns with horizontal moves. Because these orthogonal steps are strictly necessary for any valid physical sequence of moves, $h_3(S) \leq h^*(S)$ holds for all states S, proving admissibility. An exhaustive check against the exact BFS distances of all 181,440 states found 0 states where h_3 (or h_1, h_2) overestimates.

For consistency, consider any transition $S \to S'$ via a single action with unit cost $c(S, a, S') = 1$. The Manhattan component satisfies $|h_2(S) - h_2(S')| \leq 1$. Moving a single tile changes $LC(S)$ by at most 1, and only when that tile leaves or enters its goal row (or column); leaving its goal line increases the Manhattan component by 1 while removing at most one detour, and entering it decreases the Manhattan component by 1 while adding at most one detour. In every case $h_3(S) \leq h_3(S') + 1$, satisfying the triangle inequality:

$$h_3(S) \leq c(S, a, S') + h_3(S') \tag{5}$$

We also verified this exhaustively: across all 181,440 states and all 483,840 directed moves between them, h_3 never drops by more than 1, i.e. there are 0 inconsistent edges. Thus, h_3 is monotone and consistent. □

Theorem 2 (Dominance Hierarchy)**.** *For all states S, $h_1(S) \leq h_2(S) \leq h_3(S) \leq h^*(S)$.*

Proof. Every misplaced tile must move at least one step to reach its goal slot, ensuring $h_1(S) \leq h_2(S)$. Since $LC(S) \geq 0$, $h_2(S) \leq h_3(S)$ holds trivially. By Theorem 1, $h_3(S) \leq h^*(S)$. Therefore, h_3 dominates h_2, which in turn dominates h_1. The dominance is not strict: $h_3 = h_2$ on 101,400 of the 181,440 states (those with no linear conflicts). □

3 Experimental Methodology

Experiments were executed on an automated test suite across a fixed set of 38 8-puzzle configurations: 5 states sampled at each optimal depth $d \in \{4, 8, 12, 16, 20, 24, 28\}$,both states of the maximum depth $d = 31$ and the handout state ($d = 26$). For each start state, we systematically varied:

- **Heuristic Choice:** $h \in \{h_1, h_2, h_3\}$
- **WeightParameter:** $W \in \{0.0, 0.5, 1.0, 1.5, 2.0, 3.0, 5.0\}$ and greedy ($W = 0$ is run once, since it ignores h), giving 836 runs.

During every search run, four performance dimensions were recorded: nodes expanded and generated,peak memory consumption, CPU execution time (median of 3 runs) and total solution path cost g(goal), which was compared with the exact optimum d from a BFS over the full state space.

4 Results and Evaluation

4.1 Completeness

Because the state space graph of the 8-puzzle is finite ($|V| \leq 181{,}440$) and step costs are strictly positive ($c = 1$), graph search that records the best path cost found for each state is provably complete for all $W \geq 0$. Across all 836 benchmark runs, our agent achieved a 100% termination rate and every returned path was verified to start at the start state,use only legal moves and end at the goal.

4.2 Time Complexity and Node Expansions

Figure 1 depicts the mean number of generated nodes as a function of the optimal depth d for $W = 1.0$, with the uninformed search ($W = 0$) as a baseline.

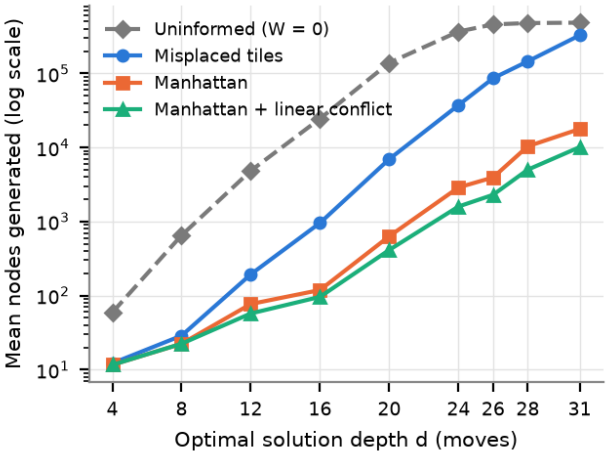


Figure 1: Mean nodes generated versus solution depth d under A^* search ($W = 1.0$) for h_1,h_2 and h_3, with the uninformed baseline ($W = 0$).

The empirical scaling curves clearly reflect the theoretical dominance relationships:

- Uninformed search ($W = 0$) degrades rapidly due to blind exploration along g-contours.It still solves every benchmark state (the space is small),but for $d \geq 26$ it generates over 4.5×10^5 nodes and at $d = 31$ expands 181,438 of the 181,440 states.
- h_1 (Misplaced Tiles) mitigates blind exploration but exhibits wide search trees at depths $d \geq 20$, generating about 7,000 nodes at $d = 20$, 37,000 at $d = 24$ and 330,000 at $d = 31$.
- h_2 (Manhattan Distance) provides an order-of-magnitude reduction in generated nodes (13–22× fewer than h_1 for $d \geq 24$).
- h_3 (Linear Conflict) achieves the steepest pruning, cutting node expansions by an additional 42–52% compared to h_2 at higher depths ($d \geq 24$).

Figure 2 demonstrates the effect of scaling W on nodes generated for the 13 hardest states ($d \geq 24$). At

$W = 0.5$ the search is still close to Dijkstra-like uniform spreading; as W increases to 2.0 it moves towards greedy best-first behavior,cutting node generation by two orders of magnitude for h_2 and h_3 (e.g. h_3:about 120,600 at $W = 0.5$, 4,300 at $W = 1$ and 570 at $W = 2$). Beyond $W = 2$, the reduction for h_2 and h_3 flattens,exhibiting diminishing returns,while the weaker h_1 keeps improving steadily up to greedy search.

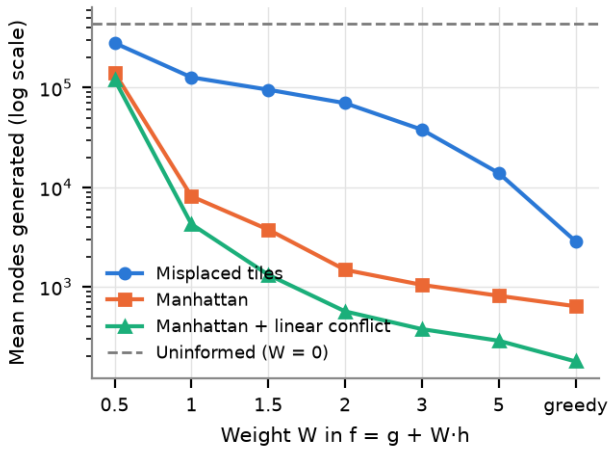


Figure 2: Mean nodes generated versus heuristic weight W across different heuristic functions on the 13 states with $d \geq 24$; the dashed line is the uninformed baseline ($W = 0$).

4.3 Cost Optimality and the Upper Bound

Under $W = 1.0$, consistency guarantees that the first goal state popped from the priority queue has an optimal path cost $g = d = C^*$.Empirically,every run with $W \leq 1$ (including $W = 0.5$) returned an optimal solution.

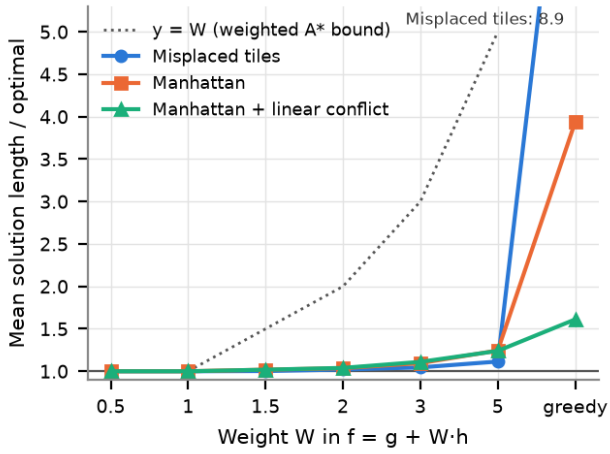


Figure 3: Mean suboptimality ratio C/C^* over all 38 states as a function of heuristic weight W; greedy search has no bound.

For $W > 1$, WA^* provides a theoretical suboptimality bound: $C \leq W \cdot C^*$.As shown in Figure 3, empirical solution costs never violate this bound and remain significantly tighter than it.The mean ratio is at most 1.02 at $W = 1.5$, 1.04 at $W = 2$ and 1.11 at $W = 3$; even at $W = 5.0$ it is only about 1.24 for h_2 and h_3 (1.12 for h_1),although individual runs reach $2.17 \cdot C^*$. Greedy search,which has no bound, averages 1.6, 3.9 and $8.9 \cdot C^*$ for h_3, h_2 and h_1 respectively.

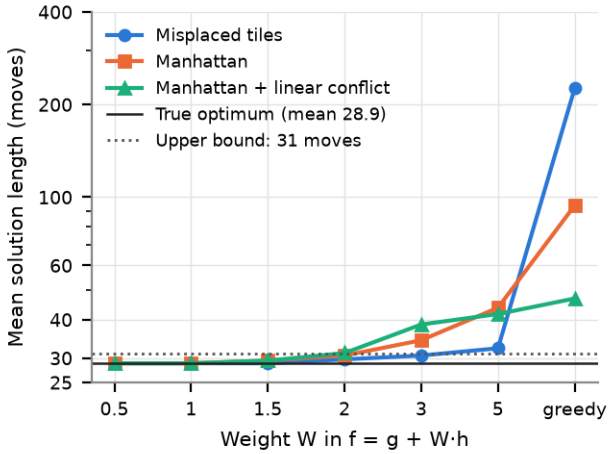


Figure 4: Mean solution path lengths on the 7 states with $d \geq 28$, compared against their true optimum and the 31-move diameter.

Figure 4 compares empirical path lengths to the theoretical diameter of the 8-puzzle state graph.The maximum optimal depth for any solvable 8-puzzle configuration is known to be 31 moves.All optimal searches ($W \leq 1.0$) strictly obey this ceiling. Weighted A^* with $W \in \{2, 3, 5\}$ mostly stays close to it:only 23 of its 342 runs exceed 35 moves.Greedy search,however frequently takes long detours: 73 of its 114 runs exceed 35 moves,with means of 47, 94 and 226 moves on the $d \geq 28$ states for h_3,h_2 and h_1, and a worst case of 381 moves with h_1.The more accurate the heuridtic, the better greedy guidance follows the true goal gradient.

4.4 Space Complexity

Because standard A^* stores the full open priority queue and every reached state in memory,its space complexity scales proportionally with the number of generated nodes. Figure 5 tracks the peak number of stored states (frontier plus explored) versus problem depth.

Heuristic quality plays a decisive role in memory viability: no run ran out of memory,but on the hardest instances ($d = 31$) h_1 stores about 145,000 states, nearly 80% of the whole state space and close to the uninformed search (all 181,440),whereas h_2 stores about 10,000 and h_3 about 5,800, retaining a compact footprint throughout the search envelope.

4.5 Heuristic Accuracy Distribution

To evaluate why h_3 outperforms h_2 and h_1,Figure 6 plots the distribution of the heuristic ratio $h(n)/h^*(n)$ across all 181,439 non-goal states.

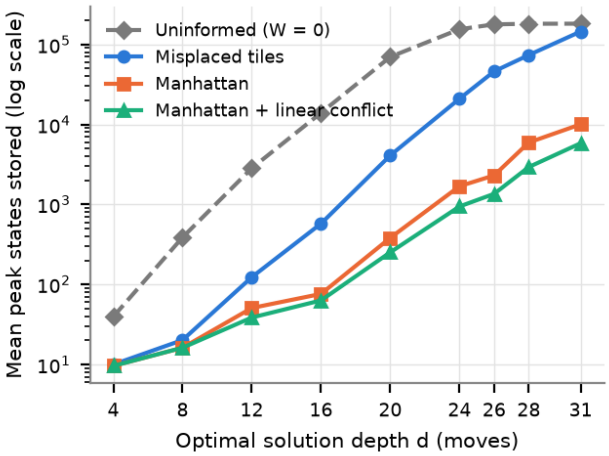


Figure 5: Peak memory footprint (mean peak number of stored states) versus problem depth d.

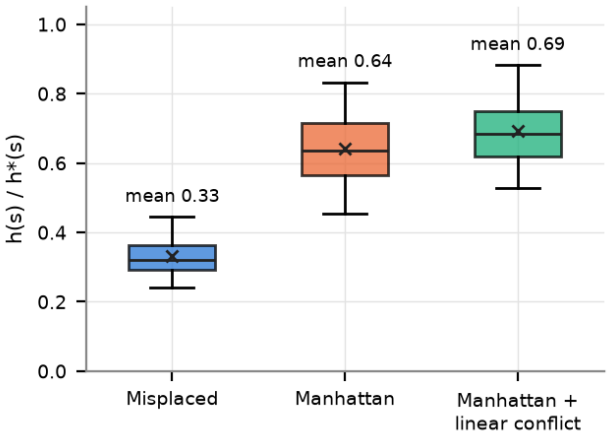


Figure 6: Boxplot distribution of heuristic accuracy ratio $h(n)/h^*(n)$ across all 181,439 non-goal states (whiskers at 5th–95th percentile, $\times$ = mean).

An ideal heuristic yields $h(n)/h^*(n) = 1.0$.While h_1 severely underestimates true remaining path distances (median ratio ≈ 0.32, mean 0.33),h_2 captures substantial distance structure (median ≈ 0.64, mean 0.64).Linear Conflict (h_3) elevates the median accuracy ratio to $\approx$ 0.68 (mean 0.69).This is a decent gain in accuracy, but because it compounds at every level of the search tree it is enough to eliminate 42–52% of the expansions on hard instances.

4.6 Solution Path Trajectory

Figure 7 illustrates the step-by-step state sequence discovered by the agent for the handout start state ($d = 26$) using h_3.A^* ($W = 1.0$) finds an optimal 26-move solution after generating 2,295 nodes,while greedy search generates only 193 nodes but returns a 46-move solution.

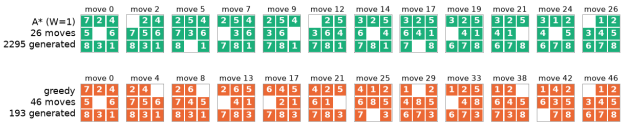


Figure 7: Sequential transition of board states for the handout state along the optimal solution path found by A^* with h_3 (top) and the path found by greedy search with h_3 (bottom).

5 Discussion and Search Principles

The empirical results illustrate several fundamental principles governing heuristic search:

1. **The Power of Information and Dominance:** Moving from h_1 to h_3 compresses the effective branching factor b^* on hard instances ($21 \leq d \leq 31$) from ≈ 1.47 down to ≈ 1.28 (uninformed search: ≈ 1.57).In exponential search trees, even minor reductions in b^* yield orders-of-magnitude savings in runtime and memory at scale.

2. **Accounting for Interaction Constraints:** Relaxed model heuristics (such as Manhattan distance) treat components independently.Linear conflict demonstrates that modeling even simple subgoal interactions within a line improves estimate fidelity (mean h/h^* from 0.64 to 0.69) while preserving admissibility, and that such a small gain still roughly halves the search effort.

3. **The Pragmatism of Weighted A^*:** When computational resources (time or RAM) are strictly constrained, enforcing strict cost optimality ($W = 1$) is often impractical. Weighted A^* with $W \in [1.5, 2]$ provides an outstanding engineering compromise: with h_2 or h_3, $W = 1.5$ eliminates 53–67% of node expansions and $W = 2$ eliminates 80–84%, while the mean path cost rises by less than 2% and 4% respectively.

6 Conclusion

We have implemented and evaluated generalized best-first search for the 8-puzzle problem under varying heuristic formulations and weights.We have proved that the Linear Conflict heuristic h_3 is admissible and consistent and it has established its dominance over Misplaced Tiles and Manhattan Distance.Experiments validate that stronger heuristics drastically reduce search space volume,and that weight parameter tuning enables control over the trade-off between execution speed and path optimality.